

SEINAJOKI, Finland

WMO: 029245

Lat: 62.692N

Lon: 22.832E

Elev: 92

StdP: 100.23

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-22.1	-28.1	0.3	-25.6	-24.2	0.4	-22.0	9.2	-4.9	8.4	-5.0	0.8	130	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	27.8	18.4	25.2	17.3	23.2	16.4	19.8	24.8	18.6	23.3	17.4	22.0	3.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.9	16.9	12.2	20.7	15.8	11.3	19.3	56.9	25.1	52.6	23.4	49.3	22.0	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	7.0	6.2	-28.6	28.7	3.0	3.1	-30.8	31.0	-32.5	32.8	-34.2	34.5	-36.3	36.8		
			-28.7	20.5	2.7	1.9	-30.6	21.9	-32.2	23.0	-33.7	24.1	-35.7	25.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-7.2	-6.6	-3.6	3.1	8.9	13.0	16.2	14.4	9.8	3.8	-0.2	-3.6
	DBStd	9.33	7.10	6.88	5.70	3.34	4.23	3.51	3.37	3.10	3.47	4.08	4.79	6.01
	HDD10.0	2679	532	464	423	207	72	10	0	3	46	193	306	423
	HDD18.3	5243	790	698	681	456	295	166	85	128	257	450	556	681
	CDD10.0	508	0	0	0	1	36	99	193	138	39	2	0	0
	CDD18.3	31	0	0	0	0	1	4	19	6	0	0	0	0
	CDH23.3	438	0	0	0	0	26	69	280	63	0	0	0	0
	CDH26.7	88	0	0	0	0	2	14	62	9	0	0	0	0
Wind	WSAvg	2.7	2.8	2.8	3.0	2.8	2.9	2.9	2.4	2.3	2.5	2.6	2.8	2.9
Precipitation	PrecAvg	559	37	27	27	24	42	60	83	66	52	57	45	39
	PrecMax	669	67	51	53	79	78	102	134	113	133	120	101	64
	PrecMin	462	9	1	5	1	9	21	20	19	20	16	6	12
	PrecStd	66	16	15	12	18	22	22	36	26	31	28	23	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	6.0	11.0	19.8	26.3	28.9	30.9	28.0	21.1	14.2	10.2	6.8
		MCWB	4.1	3.6	5.5	10.7	15.3	18.2	19.7	18.6	15.2	11.6	9.4	6.4
	2%	DB	3.1	3.9	7.1	15.2	23.1	25.1	28.8	25.0	19.0	12.1	8.2	5.1
		MCWB	2.6	2.7	3.8	8.7	14.3	16.7	19.4	17.2	14.6	10.6	7.7	4.5
	5%	DB	2.1	2.2	5.1	12.8	20.2	22.2	26.8	23.0	17.2	11.0	7.1	4.1
		MCWB	1.7	1.6	2.5	6.9	12.8	15.1	18.1	16.6	13.6	9.7	6.6	3.4
	10%	DB	1.0	1.1	3.2	10.1	17.8	20.2	24.8	20.9	15.9	9.9	5.9	3.0
		MCWB	0.8	0.7	1.5	5.8	11.3	14.4	17.6	16.0	13.1	9.0	5.5	2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	3.9	6.4	11.7	16.7	19.7	21.7	21.0	16.5	12.4	9.5	6.1
		MCDB	4.4	5.0	9.9	17.2	23.1	24.6	27.9	25.4	19.4	13.9	10.2	6.3
	2%	WB	2.6	2.6	4.2	9.3	15.2	17.8	20.4	18.7	15.3	11.1	7.8	4.6
		MCDB	2.8	3.3	6.4	14.4	22.0	23.9	26.2	22.4	17.9	12.1	8.2	4.9
	5%	WB	1.5	1.7	2.9	7.6	13.7	16.3	19.3	17.5	14.3	10.1	6.4	3.3
		MCDB	1.8	2.2	4.6	11.5	19.1	21.2	24.4	21.0	16.5	10.8	6.8	3.7
	10%	WB	0.6	0.8	1.7	5.9	11.8	15.1	18.2	16.4	13.3	8.8	5.2	2.3
		MCDB	0.8	1.1	2.9	9.7	16.7	19.2	23.1	19.7	15.4	9.4	5.4	2.6
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.8	9.1	9.4	11.6	10.9	11.4	10.7	9.2	6.1	4.3	5.1
		MCDBR	4.6	5.5	9.3	15.2	16.5	14.9	15.6	14.3	11.6	7.2	4.7	4.3
		MCWBR	4.5	4.8	6.5	9.1	8.5	7.7	7.2	7.7	7.6	5.7	4.4	4.2
	5% WB	MCDBR	4.4	5.2	8.4	13.6	14.0	13.1	12.8	11.6	10.0	6.1	4.6	4.3
		MCWBR	4.3	4.6	6.1	8.7	7.7	7.2	6.5	6.8	6.9	5.3	4.4	4.2
Clear-Sky Solar Irradiance	taub	0.236	0.291	0.316	0.356	0.352	0.350	0.373	0.358	0.334	0.316	0.257	0.352	
	taud	2.061	2.181	2.268	2.343	2.429	2.455	2.412	2.445	2.507	2.393	2.147	2.597	
	Ebn at Noon	411	647	784	827	866	873	839	820	774	629	382	243	
	Edh at Noon	33	64	86	99	99	99	101	89	70	53	30	19	
All-Sky Solar Radiation	RadAvg	0.18	0.75	2.26	3.69	4.98	5.38	5.05	3.86	2.28	0.91	0.23	0.09	
	RadStd	0.03	0.17	0.23	0.43	0.53	0.47	0.49	0.41	0.26	0.18	0.04	0.01	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (44 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page